

Subham Satapathy

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SUMMARY

Software Engineer with over 3 years of experience building cloud-scale distributed systems, Spark/Scala pipelines, and production observability. Delivered 4× runtime reduction on billion-scale workloads and built end-to-end monitoring (Datadog + Terraform). Built agentic AI automation (LLM-orchestrated workflows) using instrumentation, guardrails, and iteration. Strong fundamentals in architecture, testing, CI/CD, debugging, and live-site triage.

TECHNICAL SKILLS

- **Programming:** Python, Scala, Java, C++, JavaScript, Swift, Objective-C
- **Distributed Systems & Data:** Apache Spark, Apache Iceberg, Batch Pipelines, Redis Streams, Kafka, Elasticsearch
- **AI/Agentic Systems:** RAG Pipelines, MCP, A2A, Embeddings, Vector Databases, Retrieval, Tool Calling, Guardrails
- **Frameworks & APIs:** Pytorch, Tensorflow, LangGraph, LangChain, Node.js, Express.js, Spring Boot, REST APIs, GraphQL
- **Cloud & Databases:** AWS (EC2, EKS, ECS, S3, RDS, VPC, IAM), Kubernetes, Docker; PostgreSQL, MySQL, MongoDB
- **CI/CD & Observability:** Buildkite, GitHub Actions, Datadog, CloudWatch, Terraform (IaC)

PROFESSIONAL EXPERIENCE

Software Engineer, Rokt

July 2025 - Present | San Francisco, CA

- Reduced identity graph stats pipeline runtime 4h → 1h by optimizing Spark/Scala execution (checkpointing, partition strategy and memory-safe DataFrame design) for billion-scale processing.
- Architected an end-to-end graph vitals monitoring system computing 10+ statistical metrics (percentiles, distributions, time-window trends), persisting outputs to Apache Iceberg for reliable downstream analytics.
- Provisioned production-grade Spark observability using Datadog dashboards/monitors + Terraform (IaC), launching alerts for runtime regression, failure rate, executor OOMs, shuffle spill, and data freshness across mission-critical pipelines.
- Automated a content-generation workflow by prototyping an LLM-driven agent in n8n that orchestrates prompting and enrichment steps, cutting drafting turnaround from multi-hour effort to sub-10-minute runs.
- Improved the scalability and reliability of backend services handling large volumes of unstructured document inputs, laying the foundation for search, document understanding, and applied AI workflows.

AI Engineer, AI Growth Agent

May 2025 - July 2025 | San Francisco, CA

- Implemented context-aware keyword research logic that mines search-volume, intent and competitor gaps to surface high-value terms, increasing targeted keyword coverage for new content by 3×.
- Generated SEO assets JSON-LD schema, meta titles, slugs and alt text directly in the flow, ensuring 100 % of published articles meet Google structured-data guidelines and qualify for rich-result snippets.
- Instrumented robust monitoring & failure logging that streams real-time pipeline errors to Slack, cutting mean time to recovery from content failures by more than half.
- Built infrastructure for document ingestion, preprocessing, parsing, and workflow orchestration for cloud-based intelligent document features, improving reliability and turnaround time of downstream processing.
- Developed tooling to inspect unstructured-document processing outputs, validate extracted content and metadata, and accelerate iteration on retrieval-oriented workflows.

Software Engineer, Nix, Inc.

Jun 2024 - May 2025 | San Jose, CA

- Built infrastructure to support document ingestion, preprocessing, and workflow orchestration for cloud-based intelligent document features, improving turnaround time and reliability of downstream processing.
- Developed visualization and operational tooling that helped teams inspect document-processing outputs, monitor system behavior, and quickly prototype workflow improvements.
- Improved the reliability and scalability of backend services handling unstructured document inputs, building foundations useful for applied AI and data-driven product workflows.
- Automated CI/CD workflows with Docker, Kubernetes, and GitHub Actions, increasing deployment speed and reducing errors while enabling rapid, high-quality feature releases in AWS's production cloud environments.
- Integrated third-party APIs and built robust REST endpoints, expanding Nix's platform ecosystem, accelerating strategic partner integrations, and enabling innovative document workflow enhancements for global enterprise clients.

Software Engineer, Motorola Solutions

Jan 2022 - Jul 2023 | Bangalore, India

- Worked on application infrastructure supporting real-time audio, camera, and device-signal inputs, improving stability and performance for multimodal, latency sensitive mobile workflows.
- Improved memory usage and resource management for media-heavy iOS features, enabling more reliable performance under demanding real-world operating conditions.
- Built debugging and validation workflows that accelerated iteration on complex communication features involving

streaming, device state, and secure data handling.

- Coordinated with QA and release teams on test planning and execution, contributing to a 20% reduction in post-release bugs and consistently improving software quality across multiple development and release cycles.
- Optimized iOS app memory usage and resource management, boosting battery efficiency and reliability for users in demanding public safety environments and extended operational deployments.

Research Intern, Indian Institute of Science

July 2021 - Jan 2022 | Bangalore, India

- Designed and trained a hybrid HMM-DNN Automatic Speech Recognition model for the Kannada language from scratch, significantly advancing speech tech for low-resource regional dialects.
- Led data validation and preprocessing efforts across 10,000+ samples to ensure linguistic consistency, completeness, and formatting integrity, enabling high-quality model training and evaluation.
- Contributed to securing research funding from the Bill & Melinda Gates Foundation by demonstrating the project's potential impact on inclusive voice-based technologies for underserved populations

Software Engineer Intern, Samsung R&D

May 2021 – July 2021 | Bangalore, India

- Implemented a high-performance Smoothed Particle Hydrodynamics (SPH) simulation in C++ using GPU compute shaders and OpenGL, enabling efficient modeling of fluid dynamics in real time.
- Achieved rendering of 200,000+ particles at 30 FPS within a custom graphics pipeline, showcasing GPU-accelerated computation for resource-constrained environments.

EDUCATION

Master of Engineering, University of Maryland

College Park, MD

Software Engineering

3.97/4

Bachelors of Engineering, RV College of Engineering

Bangalore, India

Electronics and Communication Engineering

3.89/4

PROJECTS

AWS-Driven Container Orchestration and CI/CD Implementation | AWS, Docker Compose, Kubernetes, GitHub Actions

- Designed and deployed a fully containerized application on AWS Elastic Kubernetes Service (EKS), implementing Ingress, Load Balancing, and Horizontal Pod Autoscaler (HPA) for dynamic scalability and optimal traffic management.
- Automated CI/CD pipelines using GitHub Actions, streamlining code integration, testing, and deployment with Docker images pushed to Amazon Elastic Container Registry (ECR) and deployed to Kubernetes clusters.
- Implemented comprehensive monitoring and logging with Amazon CloudWatch, enhancing system performance insights and security through environment variable management, encryption, and secure API routing.

E-Learning Management System | SpringBoot, Java, JWT, Heroku, Javascript, MySQL, AWS EC2

- Developed and deployed a scalable backend using SpringBoot and Java, with secure JWT-based authentication, MySQL database management, and implemented security best practices including HTTPS, data encryption, and input validation.
- Built dynamic, responsive UI components with JavaScript, integrated them with backend services via REST APIs, and deployed the application on Heroku and AWS EC2, leveraging cloud resources for scalability, security, and performance.

Improving Hate Speech Detection using Large Language Models | Pytorch, NLP, LLM, MLM

- Drove 25x size reduction and 30x faster prediction for hate speech detection by employing optimization strategies including knowledge distillation, pre-training, and fine-tuning of LLMs without compromising accuracy.
- Balanced performance, efficiency, and generalization of hate speech models by leveraging strategies like MLM, distillation and multi-task learning across diverse datasets achieving similar accuracy to state-of-the-art models.

Speech Emotion Recognition | Pytorch, Python, Librosa

- Extracted Mel-Frequency Cepstral Coefficients (MFCC) features from speech data using Librosa to represent emotional tone and pitch variations effectively.
- Designed and trained a multi-layer perceptron (MLP) model using PyTorch, achieving 88% accuracy in classifying eight distinct emotions across the TESS and RAVDESS datasets.

CERTIFICATIONS

[Gold Medal for Programming in Java by IIT, Kharagpur](#)

[Multi AI Agent Systems with crewAI, DeepLearning.AI](#)

[MCP: Build Rich-Context AI Apps with Anthropic, DeepLearning.AI](#)

[Ethical Hacking: Footprinting and Reconnaissance, LinkedIn Learning](#)

[Objective-C Essential Training, LinkedIn Learning](#)